Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/24/2005
PI Name:	Koscheyev, Victor S Ph.D.		
Project Title:	Development of a Finger Calorimeter to Indicate General Thermal Status of Astronauts Onboard and During EVA		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	55455	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
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No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 9 1521		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A specific aim of this project is to design and test a miniaturized heat flux/temperature finger sensor and monitoring system to function as a finger calorimeter, applying experimental data collected in rest and exercise laboratory conditions. Years 1-2 have been devoted to developing and testing different heat flux ring prototypes. In collaboration with the Mini-Mitter, Inc. company, a telemetric monitoring system for measuring finger heat flux/temperature is in the process of development and testing, evaluating several variations of heat flux rings. A telemetric finger temperature sensor has been tested with human subjects in different thermal environmental conditions, under varying heat deficit/surpluses in the body. Mini-Mitter is working on the completion of the heat flux component of the finger calorimeter that will also be tested and evaluated in laboratory conditions. In collaboration with the NASA JSC EC/EVA laboratory, Year 3 will focus on adjustment of the physiological range of the sensor for compatibility with space equipment, improvement of the sensor and monitoring system, adaptation of the finger calorimeter to the space suit, and ground-based testing. In addition, initial efforts are underway to assess the efficacy of		

	physiologically/anatomically designed warming glove prototypes. Develop a miniaturized heat flux/temperature finger sensor and monitoring system for eventual use in evaluating thermal status of astronauts during EVA. Our prior research demonstrated that temperature/heat flux density from the finger is highly correlated with heat content in the body in the "vasomotor zone". This finding is the basic rationale for the development of a finger calorimeter. The heat flux/temperature finger sensor and monitoring system will be a key component in the advanced EVA development of an automatic thermal feedback system to control thermal status of astronauts during EVA by maintaining thermal balance through the life support system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Applications of the technology for better monitoring of thermal status of personnel in other types of extreme conditions, such as military, HAZMAT situations. Advances in designing support technologies for human protection.
Task Progress:	In FY 2004, activities focused on 1. physiological studies to set the parameters in which finger temperature/heat flux will be informative in determining overall body thermal status as monitored by a heat flux/temperature ring on the finger; 2. assessment of a heat flux ring prototype under the currently used Phase 6 EVA glove; 3. pilot studies on the efficacy of different physiologically-based glove designs to provide hand comfort under significant chilling of the rest of the body; 4. additional studies on the effectiveness of a head/hand bypass in the tubing of our recently developed shortened liquid cooling/warming garment (SLCWG) in prolonging hand comfort during body chilling; 5. testing of the Mini-Mitter VitalSense monitor, transmitter and skin temperature sensors in the laboratory; 6. collaboration with Mini-Mitter on the development and refinement of the finger calorimeter. 1. Continuation of physiological studies and statistical analyses of the full data set to designate the parameters in which finger temperature/heat flux will be informative in determining overall body thermal status as monitored by a heat flux/temperature ring on the finger. 2. 16 experiments were conducted to obtain additional physiological information about heat flux intensities and thermal status of the body. A heat flux ring prototype was tested under the currently used Phase 6 EVA glove. Male and female subjects were assessed under two shortened liquid cooling/warming garment (SLCWG) conditions, mild cooling with inlet water temperature at 20ºC; severe cooling with inlet water temperature at 8ºC. The 16 experiments measured the density of heat flux through the glove thickness compared with the uncovered hands, the specific location of the ring on the finger, and the possibility to register heat flux if the sensor is placed external to the pressurized rubber bladder. We also compared signal intensity with sensors placed within or outside the Phase 6 pressurized glove. 3. The efficacy of two different physiologically-based warming glove designs to provide hand comfort under significant chilling of the rest of the body was assessed in a total of 14 experiments. The prototypes consisted of 1. shortened gloves with medial and distal phalanges uncovered; 2. glove with tubing covering the main vessel pads of alternating fingers (thumb, 3rd, 5th fingers), and a control condition with the hand uncovered. In both warming glove prototype conditions, it was possible to extend the time interval before the experience of finger discomfort. After 134 minutes, finger temperature (Tfing) in both warming gloves did not reach the temperature discomfort criterion set at 25oC, while the uncovered hand reached the discomfort criterion in approximately 100 minutes. These findings indicate that the physiological design of gloves is a significant advance to conserve heat on the distal part of the extremities and provide comfort. However, variations in the tubing net in the warming gloves have differential efficacy because of their particular contact with the blood vessel network. 4. The effectiveness of a head/hand bypass in the tubing of the SLCWG in maintaining hand temperature was studied by an additional 8 experiments that indicated that it was possible to significantly delay the decline in finger temperature to the 25oC finger temperature criterion of discomfort. These findings demonstrate the potential of combining this biological head/hand heat transfer with electrical heating to minimize energy consumption and prolong finger comfort during EVA. 5. Continuation of testing of the Mini-Mitter VitalSense monitor, transmitter and skin temperature sensors in the laboratory with subjects in Phase 6 gloves with and without TMG cover to evaluate signal quality and physical distance in which the signal is registered. 6. Mini-Mitter has made progress in two areas of the design of the heat flux/temperature sensor. The temperature sensor has been designed and tested in their laboratory, providing feedback to the company for improvements. The heat flux sensor is still under development and integration to the finger calorimeter prototype developed by Mini-Mitter. • Temperature sensor – Mini-Mitter has adapted its temperature patch sensor to accommodate a miniature, remote thermister that can be attached to the finger. The main sensor/transmitter is located in a pod that can be attached to the wrist, using a watch wristband. The transmitter is activated in the normal manner, using the IR port of the VitalSense monitor. After activation, the sensor will transmit for up to 10 days on a new set of two 319 hearing aid batteries. The device can be reset by removing the batteries. • Heat flux sensor - Several sample heat-flux sensors were obtained from Concept Engineering. The most sensitive of these had a transfer constant of 268.38 W/m2/mV (or 3.7261 uV/(W/m2)). All sensors had internal impedances between 50 and 60 ohms. The expected range of heat-flux measurements was stated by the University of Minnesota researchers to be 0 ñ 200W/m2 over a temperature range of 108C to 368C. Desired resolution was 10 W/m2. • Based on the researchers' stated input range, the maximum voltage expected from the heat-flux sensor is 745.22 uV. This is a very tiny voltage, and is less than the offset voltage of many low-power, low-cost op-amps. Consequently, a high-precision "chopper-stabilized" op-amp was chosen to amplify the sensor signal. The particular device prototyped has an initial maximum offset voltage of 2uV with 30nV/8C drift. The main disadvantage of this device is that it draws 150uA and cannot be powered continuously. An inverting amplifier topology was then chosen so that the output could be biased at 1.4V, and the input could still be DC coupled. The DC coupling was considered important because the response time of the sensors is approximately 3 seconds. The gain was selected to be 2000V/V, so that a maximum anticipated input voltage of 745.22uV would result in an output voltage of 2.89V. • The heat flux sensor is being designed to interface to a heart-rate sensor board, which operates from a single-rail rechargeable battery. An on-board DC-DC converter supplies a constant 3V supply over the operating range of the battery. The converter output will be used to stabilize the opamp bias and power both the amplifier and the processor during measurement. A proof-of-concept design has been prototyped. Mini-Mitter is awaiting components prior to

	completing a usable prototype.
Bibliography Type:	Description: (Last Updated: 02/07/2014)
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